

Dietary patterns and breast cancer risk among women.

OBJECTIVE: Breast cancer is the most common type of cancer in women worldwide. Several studies have examined the role of single nutrients and food groups in breast cancer pathogenesis but fewer investigations have addressed the role of dietary patterns. Our main objective was to identify the relationship between major dietary patterns and breast cancer risk among Iranian women. **DESIGN:** Hospital-based case-control study. **SETTING:** Shohada Teaching Hospital, Tehran, Iran. **SUBJECTS:** Overall, 100 female patients aged 30-65 years with breast cancer and 174 female hospital controls were included in the present study. Dietary intake was assessed using a valid and reliable semi-quantitative FFQ consisting of 168 food items. **RESULTS:** Two dietary patterns were identified explaining 24·31 % of dietary variation in the study population. The 'healthy' food pattern was characterized by the consumption of vegetables, fruits, low-fat dairy products, legumes, olive and vegetable oils, fish, condiments, organ meat, poultry, pickles, soya and whole grains; while the 'unhealthy' food pattern was characterized by the consumption of soft drinks, sugars, tea and coffee, French fries and potato chips, salt, sweets and desserts, hydrogenated fats, nuts, industrial juice, refined grains, and red and processed meat. Compared with the lowest tertile, women in the highest tertile of the 'healthy' dietary pattern score had 75 % decreased risk of breast cancer (OR = 0·25, 95 % CI 0·08, 0·78), whereas women in the highest tertile of the 'unhealthy' dietary pattern had a significantly increased breast cancer risk (OR = 7·78, 95 % CI 2·31, 26·22). **CONCLUSIONS:** A healthy dietary pattern may be negatively associated with breast cancer risk, while an unhealthy dietary pattern is likely to increase the risk among Iranian women.